LEVEL 3 · FULL DAY · APEX PRECISION MANUFACTURING

Put the manufacturing agent in a production-shaped bundle.

A standalone lab for platform engineers and CoE teams. Copy `level-3-production/`, configure targets, and practice promotion.

DAB | CI | Promotion

Full day	Dev	95%	Person
Lab duration	CI deploys this only	Eval gate	Staging and prod

WHAT THE BUNDLE OWNS

The bundle owns every deployable asset.

Gold pipeline

Declarative gold tables the agent is allowed to see.

Metric views

OEE, FPY, maintenance measures the Genie space depends on.

Genie space

Serialized agent, instructions, and table list.

Jobs

Post-deploy tests and optional Workbench paths.

App

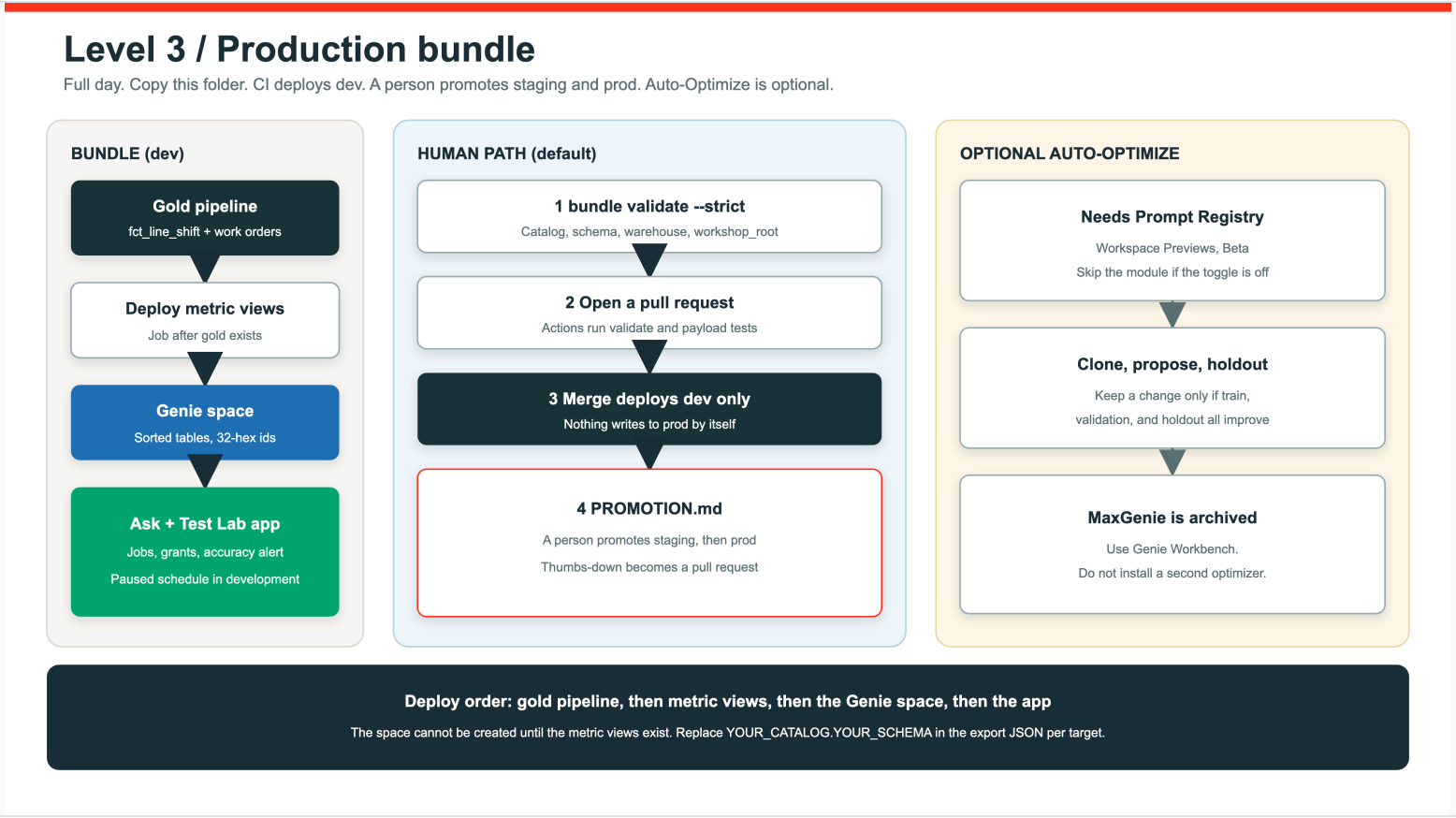
Ask + Test Lab with `source_code_path` under the bundle files root.

Permissions

Grants as code. Do not click them in by hand for the class path.

DEPLOY ORDER

Deployment order matters because each resource has dependencies.



VALIDATE FIRST

Strict validation catches broken paths and payloads before deployment.

Resource paths

YAML paths resolve relative to `resources/`.

File vs notebook

Libraries must match the actual source type.

Sorted tables

Genie table lists must be stable and complete.

32-hex IDs

Genie identifiers must be UUIDs, not display names.

`databricks bundle validate --strict --target dev` before any deploy.

CI VS PROMOTION

CI deploys dev only. A person promotes staging and production.

1	2	3	4	5
Pull request	Validate + payload tests	Merge deploys dev	Human review	Staging then prod

Lock. GitHub Actions never deploy staging or production. Follow `PROMOTION.md`.

PRODUCTION GATES

Production gates combine tests, governance, and health checks.

Payload tests

Broken DAB nodes fail in CI, not in the plant standup.

IQ Scan floor

Run the 12-check scan in Workbench before promote.

95% eval

Benchmark threshold. Below that, do not promote.

/health

App must answer before you call the deploy done.

Row filters and masks are a CoE choice. They do not mix with materialized metric views.

OPERATE

Operations need a runbook, not just a deployment.

Rollback

Previous bundle revision plus the serialized Genie file.

Principals

Staging and prod as a service principal, not a personal user.

New plant

Onboard through the model and grants, not a one-off space.

Incidents

Triage with the scorecard and `RUNBOOK.md`.

OPTIONAL OPTIMIZER

Auto-Optimize is optional and stays on staging.

Default: human PR

A thumbs-down or a failed benchmark becomes a pull request. CI validates. A person promotes.

Optional: Auto-Optimize

Clone, train / validation / holdout. Keep a change only if the score improves. Roll back if it drops. Needs Prompt Registry (workspace Preview, Beta).

MaxGenie is archived. Use Genie Workbench. Do not install a second optimizer. Skip Auto-Optimize if Prompt Registry is off.

CLOSE THE LAB

Fork the bundle, configure targets, and promote deliberately.

1. Copy `level-3-production/` as the project root.
2. Set catalog, schema, warehouse, parent path, and workshop root.
3. Replace placeholder catalog.schema in the Genie export.
4. Run `bundle validate --strict`.
5. Deploy gold, then metric views, then Genie, then the app.

You leave with a forkable bundle and a human promotion path. Staging and prod stay a person.